

LA HW

Included exercises: 1, 5, 9, 11, 19, 21, 23, 28, 32, 36, 39-40, 43-44, 46

Exercises 1-8

In Exercises 1–8, a vector space V , a basis $\mathcal{B}$ for V , and a vector $\mathbf{u}$ in V are given. Determine the coordinate vector of $\mathbf{u}$ relative to $\mathcal{B}$.

Original Exercise 1

$$V = \mathcal{M}_{2 \times 2},$$

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \right\},$$

and $\mathbf{u}$ is the matrix $\begin{bmatrix} 1 & 2 \end{bmatrix}$

Original Exercise 5

$$V = \{\mathbf{u} \in \mathcal{R}^4 : u_1 + u_2 - u_3 - u_4 = 0\},$$

$$\mathcal{B} = \left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}, \quad \text{and} \quad \mathbf{u} = \begin{bmatrix} 6 \\ -3 \\ 2 \\ 1 \end{bmatrix}.$$

Exercises 9-16

Original Exercise 9

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$\mathcal{B} = \{e^t, e^{2t}, e^{3t}\}$, $V = \text{Span } \mathcal{B}$, and $T = D$, the derivative

Original Exercise 11

$V = \mathcal{P}_2$, $\mathcal{B} = \{1, x, x^2\}$, and

$$T(p(x)) = p(0) + 3p(1)x + p(2)x^2.$$

Original Exercise 19

Let $\mathcal{B} = \{e^t, te^t, t^2e^t\}$ be a basis for the subspace V of $\mathbb{C}^\infty$. Use the method in Example 4 to find antiderivatives of the following functions:

(a) te^t

(b) t^2e^t

Exercises 20-27

In Exercises 20–27, find the eigenvalues of T and a basis for each of the corresponding eigenspaces.

Original Exercise 21

Let T be the linear operator in Exercise 9.

Original Exercise 23

Let T be the linear operator in Exercise 11.

Exercises 28-39

In Exercises 28–39, determine whether the statements are true or false.

Original Exercise 28

ments are true or false.

Every linear operator has an eigenvalue.

Original Exercise 32

tiplication.

The inverse of an invertible linear operator on a finite-dimensional vector space can be found by computing the

Original Exercise 36

the set of skew-symmetric matrices.

If T is a linear operator on a vector space with the basis $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$, then the matrix representation of T

Original Exercise 39

the eigenvalue λ .

Let V be an n -dimensional vector space and $\mathcal{B}$ be a basis for V . Every linear operator on V is of the form $\Phi_{\mathcal{B}}^{-1} T_A \Phi_{\mathcal{B}}$ for some $n \times n$ matrix A .

Original Exercise 40

Let T be the operator $D^2 + D$, where D is the derivative operator on C^∞ .

- (a) Show that 1 and e^{-t} lie in the null space of T .
- (b) Show that for any real number a , the function e^{at} is an eigenvector of T corresponding to the eigenvalue

Original Exercise 43

sides of this equation, and show that $A = \lambda^2 A$.

Let P denote the set of positive integers, and let $E: \mathcal{F}(P) \rightarrow \mathcal{F}(P)$ be defined by $E(f)(n) = f(n+1)$.

- (a) Prove that E is a linear operator on $\mathcal{F}(P)$.

Original Exercise 44

of $E^2 - E$ with corresponding eigenvalue 1.

Let $\mathcal{B}$ be the basis for $\mathcal{M}_{2 \times 2}$ given in Example 9, and let $T: \mathcal{M}_{2 \times 2} \rightarrow \mathcal{M}_{2 \times 2}$ be defined by

$$T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = \begin{bmatrix} b & a+c \\ 0 & d \end{bmatrix}.$$

- (a) Prove that T is linear.
- (b) Determine the matrix representation $[T]_{\mathcal{B}}$.
- (c) Find the eigenvalues of T .

Original Exercise 46

multiple of B .

Let B be an $n \times n$ matrix and $T: \mathcal{M}_{n \times n} \rightarrow \mathcal{M}_{n \times n}$ be the function defined by $T(A) = BA$.

- (a) Prove that T is linear.
- (b) Prove that T is invertible if and only if B is invertible.
- (c) Prove that a nonzero $n \times n$ matrix C is an eigenvector of T corresponding to the eigenvalue λ if and only if λ is an eigenvalue of B and each column of C lies